

AWS Server and Database Migration

AWS Management Console Approach

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Table of Contents

1. Overview and Prerequisites
2. AWS Management Console Access Setup
3. Server Migration using DataSync
4. EC2 Instance Creation and Setup
5. Database Migration using AWS DMS
6. Post-Migration Validation
7. Monitoring and Troubleshooting

1. Overview and Prerequisites

1.1 Document Purpose

This document provides step-by-step instructions for migrating on-premises servers and databases to AWS using the AWS Management Console. It is designed for administrators who prefer graphical user interface (GUI) operations over command-line interfaces.

1.2 Prerequisites

- AWS Account with appropriate IAM permissions
- AWS Management Console access (<https://console.aws.amazon.com>)
- AWS DataSync agent deployed on-premises
- VPN or AWS Direct Connect for secure connectivity
- Source servers with network connectivity to DataSync agent
- S3 bucket created in target region
- Target VPC and subnet configuration in AWS

1.3 Migration Scope

- Servers: Windows Server 2016/2019/2022, Linux (RHEL, CentOS, Ubuntu)
- Databases: MySQL 5.7+, PostgreSQL 9.6+, SQL Server 2016+, Oracle 11g+
- Application files and configuration data

2. AWS Management Console Access Setup

2.1 Login to AWS Console

8. Open browser and navigate to: <https://console.aws.amazon.com>
9. Enter your AWS Account ID or email address
10. Enter IAM username and password
11. Complete MFA (if enabled)
12. Select target region (e.g., us-east-1) from top-right corner

2.2 Required IAM Permissions

Ensure your IAM user/role has these permissions:

- `datasync:*`
- `dms:*`
- `ec2:*`
- `rds:*`
- `s3:*`
- `iam:PassRole`

3. Server Migration using DataSync

3.1 Deploy DataSync Agent

13. Navigate to: Services > Storage > DataSync
14. Click "Download Agent" button
15. Select virtualization platform (VMware, Hyper-V, KVM)
16. Download OVA or VHD file
17. Deploy VM in on-premises environment
18. Configure network and IP address
19. Note the DataSync agent IP for later use

3.2 Register DataSync Agent in AWS Console

20. Go to: DataSync > Agents
21. Click "Create Agent" button
22. Choose: Activation key from your DataSync agent
23. Enter agent IP: 10.0.1.50
24. Enter agent port: 34874 (default)
25. Add security group for agent connectivity
26. Click "Create Agent" and wait for activation

Step-by-Step Console Walkthrough:

- Agent Name: on-prem-datasync-agent-01
- Agent IP: 10.0.1.50
- Security Group: sg-datasync-access
- Region: us-east-1
- Status should change to "Available" in 5-10 minutes

3.3 Create Source Location (On-Premises)

27. Navigate to: DataSync > Locations
28. Click "Create Location" button
29. Select location type: "NFS" (for Linux/UNIX servers)
30. Fill in details:

Server hostname: 10.0.1.50 Subdirectory path: /data Select Agents: on-prem-datasync-agent-01 Mounting options: NOATIME

31. Click "Create Location"
32. Location ARN will appear: arn:aws:datasync:us-east-1:xxxxx:location/nfs/xxxxxxxxx

3.4 Create Destination Location (S3)

33. Still in Locations, click "Create Location"
34. Select location type: "Amazon S3"
35. Fill in details:

S3 Bucket: migration-bucket S3 Storage class: STANDARD Subdirectory: /server-data KMS Encryption: Optional Create IAM role: Yes (auto-create)

36. Click "Create Location"

3.5 Create DataSync Task

37. Navigate to: DataSync > Tasks

38. Click "Create Task" button

39. Configure task settings:

Source location: Select the NFS location created earlier Destination location: Select the S3 location Task name: Server-Migration-Task-001 Schedule: Run immediately (or set schedule)

40. Click "Next"

41. Configure Options:

Transfer mode: Change detected Preserve metadata: Yes Data validation: Point-in-time consistent Overwrite files: Yes

42. Click "Create Task"

43. Task will appear in list with status "Available"

3.6 Start and Monitor Task

44. Select the created task from list

45. Click "Start" button

46. Monitor progress in task details:

Files read: X Files written: Y Data scanned: Z GB Transfer speed: n MB/s Estimated time remaining: h:mm

47. Task completes with "Success" status

4. EC2 Instance Creation and Setup

4.1 Launch EC2 Instance from Console

48. Navigate to: Services > Compute > EC2
49. Click "Instances" in left sidebar
50. Click "Launch Instances" button
51. Step 1 - AMI Selection:

Choose a Linux image: Ubuntu Server 22.04 LTS Or Windows: Windows Server 2022 Base
(Select appropriate AMI based on source server OS)

52. Step 2 - Instance Type:

Instance type: m5.2xlarge (or match source server capacity) VCPU: 8 Memory: 32 GiB

53. Step 3 - Network Settings:

VPC: migration-vpc Subnet: migration-subnet-1a Auto-assign Public IP: Enable Security Group: migration-sg

54. Step 4 - Storage Configuration:

Root volume: 100 GiB (gp3) Additional volume for data: 500 GiB (gp3) Delete on termination: No (for data volume)

55. Step 5 - Key Pair:

Select or create key pair: migration-key Download and save .pem file securely

56. Click "Launch Instances"

57. Instance will appear in instances list with status "running" in ~2 minutes

4.2 Configure and Mount Data Volume

58. Connect to EC2 instance via SSH or RDP
59. For Linux (SSH):

```
chmod 400 migration-key.pem ssh -i migration-key.pem ec2-user@<EC2-Public-IP> # List
attached volumes lsblk # Format the data volume sudo mkfs -t ext4 /dev/nvme1n1 # Create
mount point sudo mkdir -p /mnt/data # Mount the volume sudo mount /dev/nvme1n1 /mnt/data
# Verify mount df -h
```

4.3 Copy Data from S3 to EC2

60. Attach IAM role to EC2 for S3 access (if not already done)
61. In EC2 instance, copy data from S3:

```
# Install AWS CLI (if needed) curl "https://awscli.amazonaws.com/awscliv2.zip" -o "awscliv2.zip"
unzip awscliv2.zip sudo ./aws/install # Sync data from S3 aws s3 sync
s3://migration-bucket/server-data /mnt/data --region us-east-1 # Verify data ls -lh /mnt/data du -
sh /mnt/data
```

4.4 Alternative: Import a Whole VM Using the Console (Step-by-Step)

DataSync (above) is best for ongoing or large-scale data transfer. If you simply want to move a whole virtual machine into AWS in one go — for example, a small lab server — you can export it from your hypervisor and import it directly as an EC2 instance. The steps below show this exactly as you would click through it, with the matching AWS CLI command shown under each step for reference.

Step 1 — Power off the source VM

Open your hypervisor manager (VirtualBox, VMware, or Hyper-V). Right-click the VM and choose to close it, then select "Power off the machine."

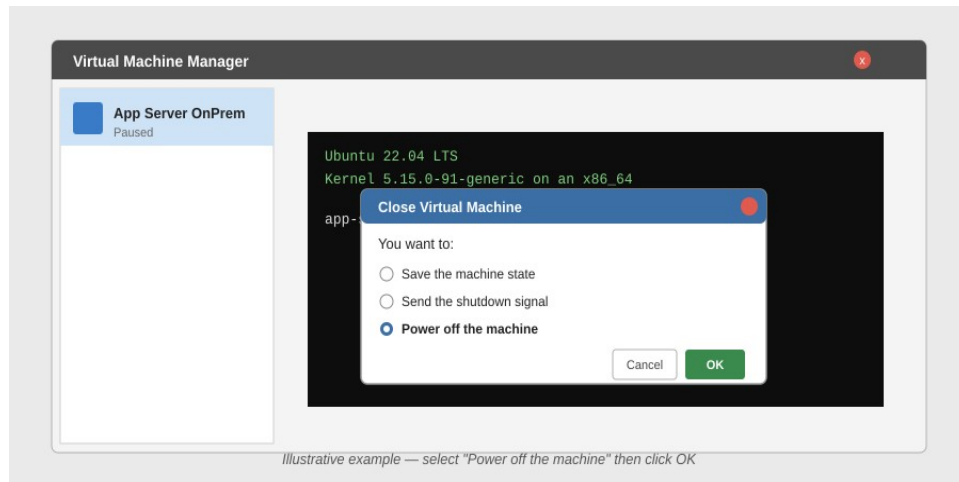


Figure 4.1 — Powering off the source VM

Step 2 — Export the VM to an OVA file

Right-click the powered-off VM and choose the export option. Pick the Open Virtualization Format (.ova) and choose a destination file on your computer.

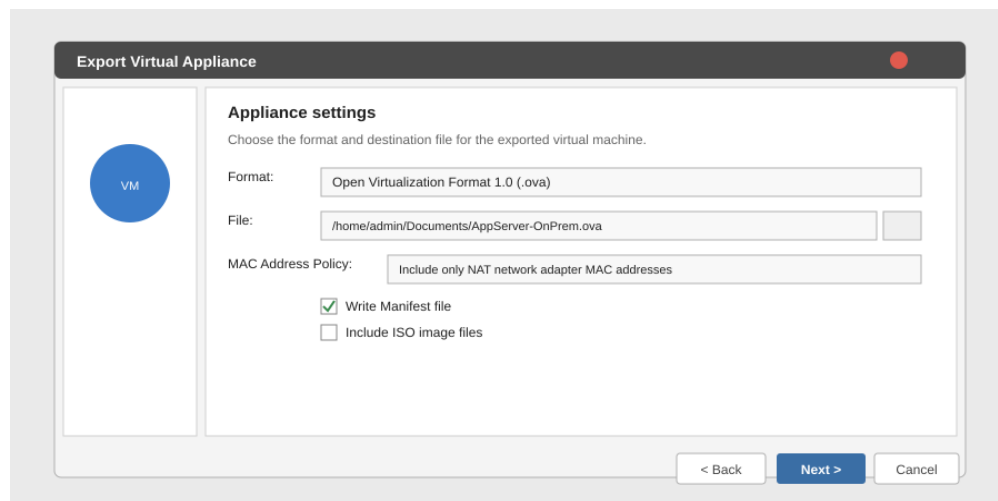


Figure 4.2 — Exporting the VM to .ova format

Step 3 — Create an S3 bucket and upload the file

62. Go to: Services > Storage > S3
63. Click "Create bucket", name it, and choose your target region

64. Open the bucket and click "Upload", then select the exported .ova file

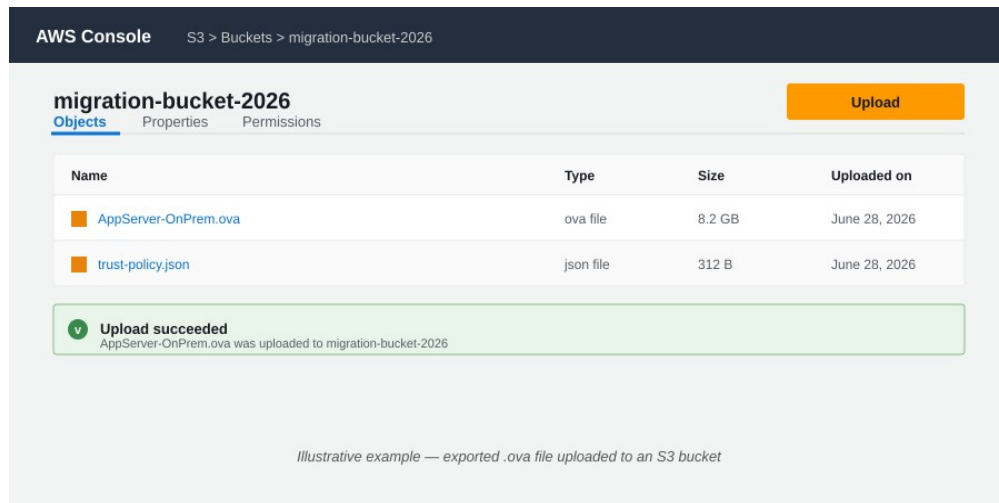
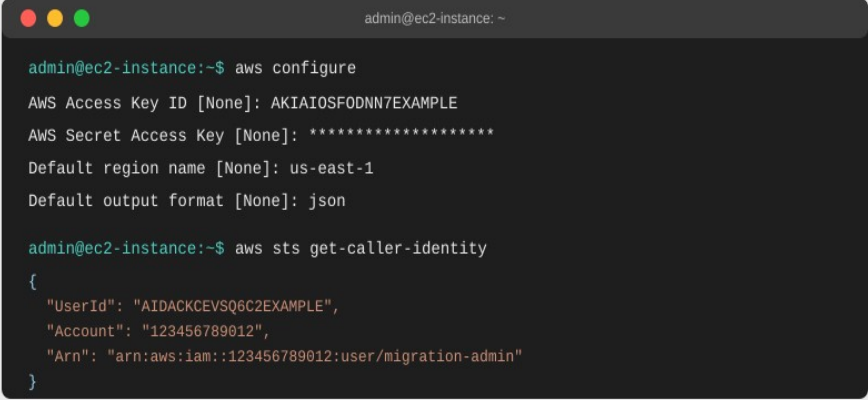


Figure 4.3 — The .ova file after a successful upload

Equivalent CLI commands: `aws s3 mb s3://migration-bucket-2026 --region us-east-1` `aws s3 cp AppServer-OnPrem.ova s3://migration-bucket-2026/`

Step 4 — Grant AWS permission to perform the import

AWS's VM import feature needs a one-time IAM role named `vmimport` with permission to read your bucket and create the image. This part is easiest done via the CLI even if you're using the Console for everything else, since there is no dedicated console page for this setup step.



```
admin@ec2-instance: ~  
admin@ec2-instance:~$ aws configure  
AWS Access Key ID [None]: AKIAIOSFODNN7EXAMPLE  
AWS Secret Access Key [None]: *****  
Default region name [None]: us-east-1  
Default output format [None]: json  
  
admin@ec2-instance:~$ aws sts get-caller-identity  
{  
  "UserId": "AIDACKCEVSQ6C2EXAMPLE",  
  "Account": "123456789012",  
  "Arn": "arn:aws:iam::123456789012:user/migration-admin"  
}
```

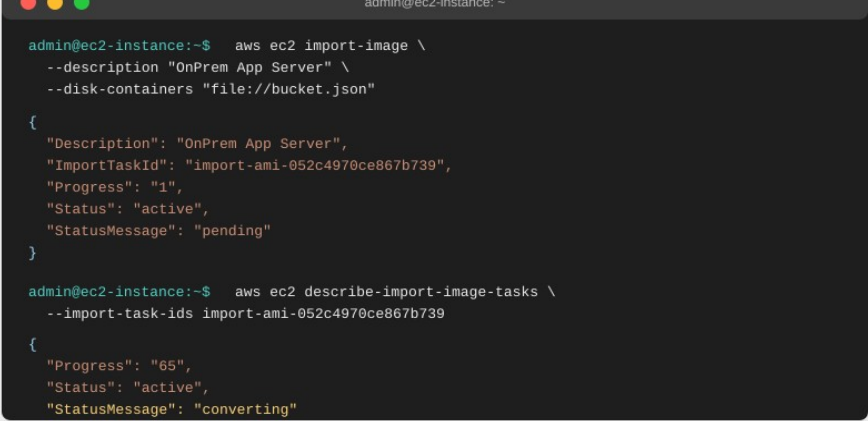
Illustrative example — values shown are sample placeholders, not real credentials

Figure 4.4 — Confirming the AWS CLI is configured and ready

```
aws iam create-role --role-name vmimport \ --assume-role-policy-document "file://trust-policy.json" \  
aws iam put-role-policy --role-name vmimport --policy-name vmimport \ --policy-document "file://role-policy.json"
```

Step 5 — Start the import and check progress

```
aws ec2 import-image --description "OnPrem App Server" \ --disk-containers "file://bucket.json"  
# Check progress any time with: aws ec2 describe-import-image-tasks --import-task-ids <task-id>
```



```
admin@ec2-instance: ~  
admin@ec2-instance:~$ aws ec2 import-image \  
  --description "OnPrem App Server" \  
  --disk-containers "file://bucket.json"  
{  
  "Description": "OnPrem App Server",  
  "ImportTaskId": "import-ami-052c4970ce867b739",  
  "Progress": "1",  
  "Status": "active",  
  "StatusMessage": "pending"  
}  
  
admin@ec2-instance:~$ aws ec2 describe-import-image-tasks \  
  --import-task-ids import-ami-052c4970ce867b739  
{  
  "Progress": "65",  
  "Status": "active",  
  "StatusMessage": "converting"  
}
```

Illustrative example — task IDs and progress values are samples

Figure 4.5 — Tracking the import task until it completes

Step 6 — Launch the imported server

65. Go to: EC2 > Images > AMIs
66. Select the newly imported AMI and click "Launch instance from AMI"
67. Complete the launch wizard exactly as in Section 4.1

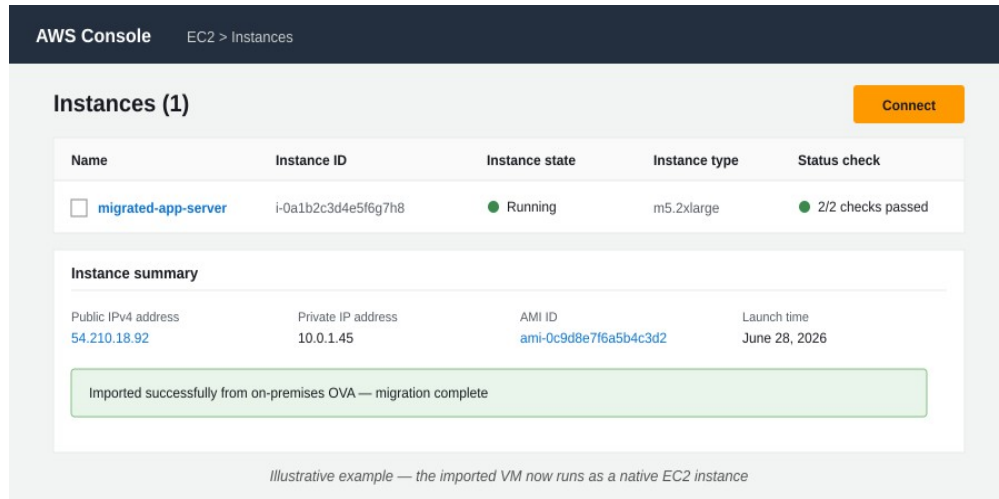


Figure 4.6 — The migrated server running as an EC2 instance

Try it yourself

68. Create a VM in VirtualBox, VMware, or Hyper-V running Ubuntu
69. Deploy a simple web application and a MySQL database on it
70. Follow Steps 1-6 above to migrate that VM into AWS

5. Database Migration using AWS DMS

5.1 Create DMS Replication Instance

71. Navigate to: Services > Migration & Transfer > Database Migration Service
72. Click "Replication Instances" in left menu
73. Click "Create Replication Instance" button
74. Fill in configuration:

Replication instance identifier: db-migration-instance-01 Instance class: dms.c5.2xlarge Engine version: Latest (3.4.x) Allocated storage: 500 GB Multi-AZ: Yes (for production) Public accessibility: No Availability Zone: us-east-1a VPC: migration-vpc Security group: dms-sg

75. Click "Create"
76. Instance creation will take 10-15 minutes. Status: "available"

5.2 Create Source Endpoint

77. In DMS console, click "Endpoints" in left menu
78. Click "Create Endpoint" button
79. Select "Source endpoint"
80. For MySQL source:

Endpoint type: Source Engine: MySQL Server name: 10.0.1.100 (source DB server IP) Port: 3306 Database name: production_db Username: admin Password: YourPassword123! SSL mode: none (or preferred for production)

81. Click "Test Endpoint Connection"
82. Select replication instance: db-migration-instance-01
83. Wait for "successful" status
84. Click "Create Endpoint"

5.3 Create Target Endpoint (RDS)

85. Click "Create Endpoint" again
86. Select "Target endpoint"
87. For RDS MySQL target:

Endpoint type: Target Engine: MySQL Server name: mydb-instance.123456789012.us-east-1.rds.amazonaws.com Port: 3306 Database name: production_db Username: admin Password: YourPassword123! SSL mode: preferred

88. Click "Test Endpoint Connection"
89. Click "Create Endpoint"

5.4 Create Migration Task

90. In DMS console, click "Tasks" in left menu
91. Click "Create Task" button
92. Configure basic settings:

Task identifier: mysql-to-rds-migration-01 Replication instance: db-migration-instance-01 Source endpoint: source-mysql-db Target endpoint: target-rds-mysql Migration type: Migrate existing data and replicate ongoing changes

93. Click "Next"
94. Select "Migrate all databases" or select specific databases

95. Configure Task Settings (Advanced):

Batch apply enabled: Yes Batch applied size: 100-1000 KB Validation enabled: Yes (for full load and CDC) Error behavior: Fail on no change LOB settings: Full LOB mode (if needed)

96. Click "Create and Start Immediately"

5.5 Monitor Migration Progress

97. Task details page shows real-time progress:

Full Load Progress: - Tables completed: X of Y - Current transfer rate: n MB/s - Elapsed time: h:mm:ss - Estimated time remaining: h:mm:ss Table Statistics: - Inserts: n - Updates: n - Deletes: n - Rows validated: n

5.6 Handle Validation Failures (if any)

98. Click "Validation" tab to view detailed results

99. For each failed table:

- Review mismatch details
- Identify root cause (LOB data, character set, triggers, etc.)
- Apply corrections on target or source
- Restart validation

6. Post-Migration Validation

6.1 Validate EC2 Instance Status

100. Go to: EC2 > Instances
101. Select your instance and verify:
 - Instance State: running
 - Instance Status Checks: 2/2 passed
 - System Status: OK
 - Instance Status: OK

6.2 Validate Database Migration

102. Connect to RDS instance using database client
103. Verify:

```
# Count tables SELECT COUNT(*) FROM information_schema.tables WHERE table_schema =  
'production_db'; # Count rows in each table SELECT table_name, table_rows FROM  
information_schema.tables WHERE table_schema = 'production_db'; # Check indexes SHOW  
INDEXES FROM production_db.users;
```

6.3 Application Testing Checklist

- Update application connection strings to point to new RDS endpoint
- Test database connectivity from application server
- Run application smoke tests
- Verify data queries return correct results
- Test write operations (INSERT, UPDATE, DELETE)
- Validate backups and snapshots
- Confirm monitoring and alerts are configured

6.4 Configure CloudWatch Monitoring

104. For EC2: Go to EC2 > Instances > Select instance > Monitoring tab
105. Enable detailed monitoring
106. For RDS: Go to RDS > Databases > Select instance > Monitoring section
107. Set up CloudWatch alarms for:
 - CPU Utilization > 80%
 - Free Memory < 20%
 - Replication Lag > 5 seconds (for read replicas)
 - Database Connections > 80% of max

7. Monitoring and Troubleshooting

7.1 Common Issues via Console

Issue	Symptoms in Console	Console Resolution Steps
Agent not responding	DataSync > Agents shows "Offline"	Check agent VM power, network connectivity, restart VM
Task failed with error	DMS > Tasks shows red status with error message	Click task > View logs, check source/target connectivity
Slow replication	DMS replication instance CPU/Memory high, transfer rate low	Increase replication instance class, optimize table mappings
RDS disk full	RDS > Databases shows storage threshold exceeded	Modify DB instance to increase allocated storage

7.2 View DMS Task Errors in Console

108. Go to: DMS > Tasks
109. Click on the task name
110. View tabs: Status, Validation, Performance, Server Objects
111. Check "Status" tab for task status and error messages
112. Review logs at: CloudWatch > Logs > /aws/dms/tasks/[task-name]

7.3 CloudWatch Dashboards

113. Create a custom dashboard for migration resources
114. Add widgets for:
 - EC2 CPU, Memory, Network metrics
 - RDS Database Connections, CPU, Storage
 - DMS Replication Instance CPU and Memory
 - S3 bucket size and data transfer metrics
 - DMS full load and CDC progress

7.4 Cutover Steps (Final Migration)

115. Schedule cutover window (typically low-traffic period)
116. Stop applications on source servers
117. Wait for DMS CDC to complete replication
118. Stop DMS task
119. Verify final data sync between source and target RDS
120. Update application DNS/connection strings
121. Start applications pointing to new RDS
122. Monitor application logs and database performance
123. Keep source systems running for 24-48 hours as rollback point